

INDEPENDENT RESEARCH PORTFOLIO / PARTICLE PHYSICS

Gravity at the Terascale: How CERN Searches for Extra Dimensions, Gravitons, and Microscopic Black Holes

Parusan Natheeswaran

Independent Research Portfolio

July 2026

*A critical review of how higher-dimensional geometry becomes a collider observable,
and why current exclusions remain model-conditional.*

Abstract

Extra dimensions become experimentally meaningful at the Large Hadron Collider only when a model specifies how higher-dimensional gravity couples to proton collisions. This critical review follows that translation from the large, flat geometry of Arkani-Hamed-Dimopoulos-Dvali models and the warped geometry of Randall-Sundrum models to three graviton signatures: missing transverse momentum from real Kaluza-Klein emission, localized spin-2 resonances, and nonresonant spectral or angular distortions from virtual exchange. It then distinguishes semiclassical high-multiplicity black-hole and string-ball searches from quantum-black-hole few-body benchmarks. Representative ATLAS and CMS Run 2 analyses, including full-data results released in 2024 and 2026, show no compelling excess. Their exclusions nevertheless constrain different quantities: a fundamental gravity scale, convention-dependent effective cutoffs, resonance masses and couplings, or production thresholds tied to generator assumptions. The paper argues that these limits cannot be combined into a universal bound on extra dimensions, especially where effective-interaction or semiclassical descriptions approach their validity limits. CERN's program is best understood as a network of complementary, model-conditional tests that progressively restricts terascale gravity while leaving the broader existence question open. Independent cosmic-ray and astrophysical arguments also show why hypothetical collider-scale black holes do not constitute a credible hazard.

Keywords: extra dimensions; Kaluza-Klein gravitons; LHC; ATLAS; CMS; microscopic black holes; terascale gravity

1. Introduction

The enormous separation between the electroweak and gravitational scales is one of the central organizing problems in particle physics. The Standard Model describes electromagnetic, weak, and strong interactions with exceptional precision, yet gravity becomes comparable in strength only near the Planck scale, around 10^{19} GeV. Protecting the much smaller electroweak scale from quantum corrections appears to require either unexplained parameter adjustment or new structure. Extra-dimensional models offer one possible structure: gravity may look weak because it propagates through a geometry that Standard Model particles do not fully inhabit [1,3].

Two influential proposals show why “extra dimensions” is not one hypothesis. Arkani-Hamed, Dimopoulos, and Dvali use a large compact spatial volume to dilute the apparent strength of gravity and permit a fundamental scale M_D near the teraelectronvolt regime [1]. Randall and Sundrum instead use a warped fifth dimension to exponentially redshift physical mass scales and predict a discrete spectrum of massive graviton excitations [3]. These constructions address the same hierarchy but produce different spectra, couplings, and collider signatures. Their theoretical motivation does not count as evidence for either geometry; it establishes a reason to build falsifiable tests.

ATLAS and CMS cannot photograph a compact dimension. They measure tracks and energy deposits from proton collisions and ask whether their distributions contain the consequences of a specified model. An escaping KK graviton would be inferred from transverse momentum imbalance against a jet or photon. An on-shell warped graviton could form a resonance in a diphoton mass spectrum. The exchange of an unresolved KK tower could modify a high-mass spectrum or the centrality of dijet events. If gravity became strongly coupled at partonic energies accessible to the LHC, models also predict energetic few-body or high-multiplicity states labelled quantum or semiclassical microscopic black holes [9,12-15].

These labels can encourage an oversimplified story in which CERN performs one search for gravitons and another for tiny black holes. The experimental program is more exacting. Each signature requires its own trigger, event reconstruction, control samples, background model, statistical test, and parameter convention. The quantities reported as M_D , M_S , A_T , resonance mass, or black-hole threshold are not interchangeable, even when all are expressed in teraelectronvolts. Near the highest collision energies, the effective theory or semiclassical approximation used to produce a signal template may also approach its validity limit [4,5,8].

This paper asks how ATLAS and CMS translate extra-dimensional models of low-scale gravity into searches for real and virtual gravitons, KK resonances, and microscopic black holes, and what their current null results actually constrain. It argues that the LHC does not test extra dimensions as one model-independent proposition. Rather, it tests three complementary graviton regimes, consisting of missing transverse momentum from real emission, resonant or nonresonant spectral changes, and virtual-exchange angular distortions, alongside distinct few-body and high-multiplicity strong-gravity benchmarks. Because every exclusion depends on geometry, coupling, convention, cutoff, production threshold, and decay assumptions, the correct conclusion is a progressively bounded map of terascale gravity, not a universal rejection of extra dimensions.

The analysis focuses on the ADD and original Randall-Sundrum frameworks, representative Run 2 ATLAS and CMS searches, and the theoretical limits needed to interpret their exclusions. It includes the March 2026 CMS dijet-angular measurement and April 2026 CMS high-multiplicity search, the most current results in the channels considered [13,14]. It does not attempt a general review of string theory, cosmology, or every compactification model. A concise final section separates the hypothetical collider signatures from safety claims, because microscopic quantum states and astrophysical black holes belong to categorically different regimes [18,19].

The organization follows the inference chain used by the experiments. Section 2 compares the geometries and scales of ADD and Randall-Sundrum models. Section 3 derives the real-emission, resonant, and virtual-exchange signatures. Sections 4 and 5 examine how those signatures become concrete graviton and microscopic-black-hole analyses. Section 6 interprets the null results and their validity conditions, before Section 7 addresses safety and public language. This structure keeps the paper's central object in view: not the speculative image of another dimension, but the disciplined conversion of geometry into measurable evidence.

2. Why Extra Dimensions Change the Gravity Scale

The collider case for extra dimensions begins with a mismatch of scales, not with the direct visibility of another direction in space. The electroweak scale, set by the Higgs vacuum expectation value and the masses of the weak bosons, is roughly sixteen orders of magnitude below the conventional four-dimensional Planck scale. Quantum corrections make that separation difficult to regard as accidental without an organizing mechanism. Large-extra-dimension and warped-extra-dimension models supply two such mechanisms. Neither is evidence that extra dimensions exist; each is a calculable proposal that turns the hierarchy problem into predictions that can be tested and, increasingly, restricted.

2.1 Large, flat dimensions: volume dilution in ADD

Arkani-Hamed, Dimopoulos, and Dvali (ADD) proposed that the fundamental scale of gravity could be close to the electroweak scale if gravity propagates through additional compact spatial dimensions while Standard Model fields remain confined to a four-dimensional subspace [1]. In this picture, gravity is not intrinsically weak. It only appears weak at distances much larger than the compactification radius because its field lines have spread through the full higher-dimensional volume before they are observed in four dimensions.

For n flat extra dimensions of a common characteristic radius R , the scale relation can be written schematically as

$$M_{Pl}^2 \sim R^n M_D^{n+2},$$

where M_{Pl} is the observed four-dimensional Planck scale and M_D is the fundamental gravitational scale in $4+n$ dimensions [1,2]. The relation is deliberately shown as a proportionality: factors involving 2π , and even the precise numerical definitions of M_{Pl} , R , and M_D , differ among conventions. Its physical content is more important than its normalization. A sufficiently large compactification volume permits M_D to be of order teraelectronvolts even though the effective four-dimensional Planck scale is enormous.

The number of dimensions changes the required radius dramatically. A single large flat dimension would need to be so large that it conflicts with observed gravitational behavior; two or more dimensions can be much smaller and therefore more difficult to probe directly [1,2]. This dependence also matters at colliders. The density of accessible Kaluza-Klein states, phase-space weights, and cross sections all depend on n . A quoted collider limit on M_D is consequently inseparable from the assumed number and geometry of dimensions.

Compactification also turns higher-dimensional momentum into an observable four-dimensional mass spectrum. A field propagating around a compact direction has quantized momentum in that direction. To a detector restricted to ordinary space, each allowed momentum appears as a distinct Kaluza-Klein (KK) excitation. In ADD models, the compactification radius can make adjacent graviton modes too closely spaced for ATLAS or CMS to resolve one by one. Each individual mode couples very weakly, but the enormous number of modes available in a high-energy collision can produce an observable inclusive rate [2,5]. This combination explains why ADD phenomenology often appears as missing transverse momentum from real emission or as a smooth, nonresonant modification from the exchange of an unresolved KK tower.

The same mechanism that makes ADD testable also prevents the LHC from being its only judge. Short-distance gravity measurements, supernova cooling, cosmology, and astrophysical gamma-ray observations can constrain portions of the parameter space, sometimes with assumptions different from those used at colliders [2,5]. Collider searches remain distinctive because they test production and interaction at microscopic distances in a controlled laboratory environment, but their exclusions should be read as one layer of a broader empirical map.

2.2 Warped geometry: scale redshift in Randall-Sundrum models

Randall and Sundrum offered a geometrically different solution. In the original two-brane construction, a single compact extra dimension is embedded in a five-dimensional spacetime with strong curvature [3]. Standard Model mass scales on the visible brane are exponentially redshifted relative to a fundamental scale on the other brane. Schematically, a mass parameter m_0 is observed as

$$m_{vis} = e^{-k r_c \pi} m_0$$

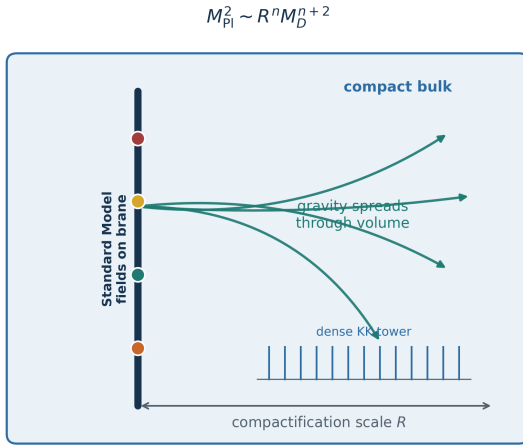
where k characterizes the curvature and r_c sets the compactification radius [3]. A moderately sized value of $k r_c$ can therefore generate a very large hierarchy without requiring the expansive flat volume used in ADD.

This distinction changes the expected collider signal. The warped geometry produces a KK graviton spectrum with discrete masses separated widely enough that the lightest states can, in suitable parameter regions, appear as resonances. Their couplings to Standard Model fields are governed not only by the resonance mass but also by a dimensionless curvature-to-Planck-scale ratio, commonly expressed as k/M_{Pl} . The coupling affects both the production rate and the intrinsic width. A weakly coupled state may form a narrow peak in a diphoton or dilepton invariant-mass spectrum; a stronger coupling broadens the state and changes the analysis response [3,11,12].

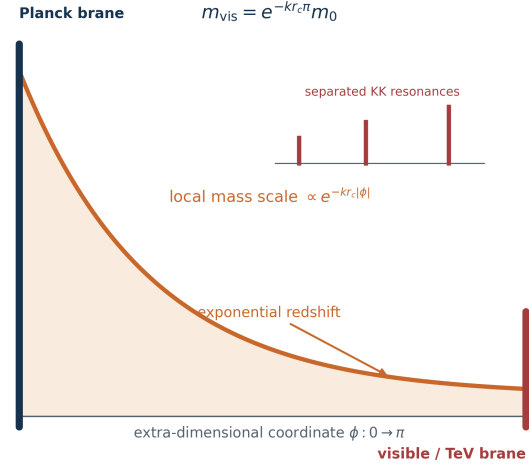
Calling both frameworks “extra dimensions” can therefore conceal more than it reveals. ADD uses a large flat volume, a quasi-continuous tower at collider resolution, and an effective gravitational scale M_D . The original Randall-Sundrum model uses warping, a discrete resonance spectrum, and mass-plus-coupling benchmarks. Even the symbol n has a different role across related analyses: an ADD result for $n=6$ is not a point on the same parameter axis as an RS1 result described as one warped dimension. The models share a higher-dimensional origin and spin-2 excitations, but they do not offer interchangeable predictions.

Two geometries, two collider phenomenologies

A. ADD: large, flat extra dimensions



B. Randall-Sundrum: warped dimension



Original schematic; geometry and spectra are not to scale.

Figure 1. Original schematic comparison of large, flat ADD geometry and warped Randall-Sundrum geometry. The compactification geometry and Kaluza-Klein spectra are not drawn to scale.

This is the first interpretive boundary of the LHC program. ATLAS and CMS cannot exclude “extra dimensions” with one mass threshold because no universal extra-dimensional signal exists. They test a chain of commitments: geometry, compactification, field localization, spectrum, coupling, and the validity of the resulting effective description. The next step is to follow that chain from KK gravitons to the detector observables on which an actual search can operate.

3. From Geometry to Collider Signatures

An extra dimension becomes experimentally meaningful only after its higher-dimensional degrees of freedom are translated into four-dimensional processes. For the models considered here, the central messenger is the graviton. Compactification replaces a continuous component of higher-dimensional momentum with a KK spectrum of massive spin-2 states [4,5]. What ATLAS or CMS observes then depends on how those states enter the hard collision: a mode can be emitted as a real particle, exchanged virtually, or produced as an on-shell resonance. The three cases share a theoretical origin but demand different observables, background models, and standards of interpretation.

3.1 Real emission: inferring an invisible recoil

In an ADD process such as $qg \rightarrow qG_{KK}$ or $q\bar{q} \rightarrow \gamma G_{KK}$, the graviton mode interacts too weakly to deposit measurable energy in the detector. Its signature is therefore not a visible track or calorimeter cluster. Because the longitudinal momentum of the colliding partons is unknown event by event, the experiment instead uses momentum conservation in the plane transverse to the beams. The vector sum of reconstructed transverse momenta should be approximately balanced in an ordinary collision; a large imbalance indicates invisible momentum. A high-energy jet or photon provides the visible recoil and triggers the event, creating the monojet or monophoton topology [4,5,9,10].

“Mono” is a useful label but an imperfect literal description. Additional jets from initial-state radiation are common, and an analysis does not identify a graviton by inspecting one dramatic event. The dominant irreducible monojet background is Standard Model production of a Z boson that decays to neutrinos. Lost-lepton W +jets events, top production, dibosons, and instrumental multijet tails also contribute. Their rates and kinematic shapes are constrained through control samples containing charged leptons or photons and through transfer factors from simulation. Sensitivity comes from a simultaneous fit across recoil or missing-momentum bins, where a new process would add a harder event population than the calibrated background expectation [9].

The monophoton channel follows the same conservation law but changes the experimental system. Photon identification suppresses much of the hadronic activity that complicates monojet events, while the smaller production rate reduces statistical reach. Relevant backgrounds include $Z(\nu\bar{\nu})\gamma$, $W\gamma$ with a lost lepton, electrons reconstructed as photons, and noncollision sources such as beam halo [10]. Monojet and monophoton results are thus complementary measurements of real emission, not repeated versions of one search. Their different partonic channels, object efficiencies, and systematic uncertainties provide independent ways to test the same model family.

3.2 Resonant production: searching for a spin-2 peak

A warped-model KK graviton can be produced on shell and decay back into Standard Model particles. In a diphoton search, the experimental object is the invariant mass

$$m_{\gamma\gamma}^2 = (p_{\gamma 1} + p_{\gamma 2})^2.$$

A narrow state would add a localized concentration of events to an otherwise smoothly falling spectrum. Diphotons are especially valuable because the electromagnetic calorimeters measure energetic photons precisely and the Standard Model continuum is smooth at high mass. The analysis estimates that continuum, accounts for jets misidentified as photons, and scans signal hypotheses across masses and widths [11,12].

A peak alone would not establish that the state was an extra-dimensional graviton. A spin-0 resonance, detector effect, or statistical fluctuation could produce a superficially similar mass feature. Attribution would require a consistent production rate, width, decay pattern, and angular distribution across channels. Because a spin-2 state has characteristic helicity structure, angular information can help distinguish it from alternatives once sufficient data exist. Cross-channel consistency in diphoton, dilepton, and possibly diboson final states would be more persuasive than the maximum local significance in any one spectrum.

3.3 Virtual exchange: broad deviations and effective scales

KK gravitons can also contribute as virtual intermediate states. The summed tower modifies Standard Model amplitudes and can interfere with them, changing the high-mass tail of a diphoton distribution or the angular structure of dijet scattering [4,5,12,13]. Unlike a narrow resonance, the effect may be spread across many bins. A search must then trust not only a local background interpolation but the predicted shape of the Standard Model process over the relevant energy range.

In the GRW treatment, virtual exchange is represented by a dimension-eight interaction built from the Standard Model energy-momentum tensor, with its strength controlled by an effective cutoff scale often denoted Λ_T [4]. Hewett and Han-Lykken-Zhang (HLZ) use related parameterizations with different normalizations, signs, and explicit dimensional dependence [5]. These conventions describe overlapping physics but do not assign the same numerical value to the suppression scale. A statement such as “extra dimensions are excluded below 13 TeV” is therefore incomplete unless it identifies the convention, process, and model assumptions.

The effective description has a second limitation. The contribution grows rapidly with partonic energy, so a collider can become most sensitive near the cutoff where the approximation itself requires scrutiny. Analyses may truncate simulated events above M_{S^*} , vary prescriptions, or report limits within an explicitly defined effective model [12]. Such measures do not invalidate the search. They define the question it answers: whether a particular low-energy representation of KK exchange, used with a stated ultraviolet treatment, is compatible with the data.

Dijet angular distributions illustrate why virtual effects need a dedicated strategy. At high invariant mass, QCD scattering is dominated by forward topologies. Contact-like new interactions tend to produce a relatively larger fraction of central events. CMS characterizes this structure using a variable such as

$$\chi_{dijet} = \exp(|y_1 - y_2|),$$

where y_1 and y_2 are the rapidities of the leading jets [13]. Small χ_{dijet} corresponds to more central, balanced rapidities. The search is consequently a shape comparison rather than a simple count above a mass threshold. Unfolding detector effects and comparing the result with high-order QCD and electroweak predictions allows the measurement to constrain broad new contributions that might never form a bump.

From higher-dimensional model to an LHC exclusion

The reported TeV scale changes meaning from row to row.

MODEL REGIME	PARTON-LEVEL IDEA	DETECTOR OBSERVABLE	ANALYSIS HANDLE	REPORTED PARAMETER
ADD real emission	jet or γ + escaping G_{KK}	large p_T^{miss} against recoil	control-region constrained recoil fit	$M_D(n)$
ADD virtual exchange	KK tower modifies $q\bar{q}, gg$ amplitudes	high-mass or central-angle distortion	tail prediction or unfolded shape	M_S or Λ_T + convention
Warped KK graviton	on-shell spin-2 state $\rightarrow \gamma\gamma$	localized invariant-mass feature	resonance scan plus width/angle tests	m_G and $k/\bar{M}_{Pl}$
Semiclassical BH / string ball	strong-gravity production well above threshold	many hard objects; S_T , multiplicity, sphericity	data-driven tail or phase-space classifier	M_{BH}^{min} + generator model
Quantum black hole	near-threshold state with few-body decay	dijet or lepton+jet high-mass feature	mass-spectrum or angular search	M_{th} + decay assumptions

A larger numerical exclusion is not automatically a stronger or more general statement.

Figure 2. Original model-to-observable map for the five collider regimes examined in this paper. Each row ends in a different reported parameter and therefore a different conditional exclusion.

These three modes establish the paper's experimental taxonomy. Real emission is inferred from recoil and missing transverse momentum; resonant production is sought as a mass-localized state with spin-dependent properties; and virtual exchange is constrained through nonresonant spectral or angular distortions. They cannot be collapsed into a single “graviton limit,” because they test different amplitudes, scales, and assumptions. The concrete ATLAS and CMS analyses make that complementarity visible.

4. How ATLAS and CMS Search for Gravitons

The practical strength of the LHC graviton program comes from analyses exposed to different uncertainties. Missing-momentum searches depend heavily on control of invisible Standard Model backgrounds. Diphoton searches depend on mass resolution and a credible description of a steeply falling continuum. Dijet angular measurements require precise shape predictions and unfolding. Agreement across these channels would be essential to a discovery claim; disagreement helps expose a mis modeled background or an overly narrow theoretical interpretation. Current null results are best understood through that network of methods rather than through a ranking of the largest excluded scale.

4.1 Monojet and monophoton recoil searches

The 2018 CMS monojet and mono- V analysis used 35.9 fb^{-1} of proton-proton collision data at $\sqrt{s}=13$ TeV [9]. Candidate monojet events were selected with large missing transverse momentum and an energetic hadronic recoil. Instead of assigning every event to a single background estimate, CMS performed a combined likelihood analysis of signal regions and lepton- or photon-enriched control regions. The visible control samples constrain the invisible $Z(\nu\nu)+jets$ contribution and related $W+jets$ processes through theoretically and experimentally calibrated transfer factors. Top, diboson, and multijet components complete the model.

That architecture matters more than the “one jet plus nothing” shorthand. Neutrinos are produced abundantly by known electroweak processes, and detector response can create apparent imbalance. A credible excess would need to persist across recoil bins after correlations among jet energy calibration, lepton efficiencies, electroweak corrections, and control-sample statistics were propagated through the fit. CMS observed no significant departure from the Standard Model expectation. Under the ADD signal assumptions used in the analysis, it excluded fundamental gravity scales M_D below 9.9 TeV for $n=2$, with the reach falling to 5.3 TeV for $n=6$, at 95% confidence level [9]. The n -dependence is part of the result: the numbers are not six measurements of one universal Planck scale.

CMS's 2017 monophoton analysis used 12.9 fb^{-1} at the same collision energy [10]. Requiring a well-identified energetic photon changes the instrumental and physics backgrounds. The irreducible $Z(\nu\nu)\gamma$ process remains, but electron-to-photon misidentification, beam-halo deposits, and photon-specific reconstruction uncertainties become important. The data again showed no significant excess. In the tested ADD models, CMS set M_D limits between 2.31 and 2.49 TeV, depending on n [10]. These values should not be read as proof that a photon search is intrinsically inferior. The analysis used an earlier and smaller data set, and each topology samples different production rates, acceptance, and uncertainties. Its value is methodological independence and a distinct observable recoil.

4.2 One spectrum, two diphoton questions

High-mass diphoton events can test both warped resonances and nonresonant ADD exchange, but the statistical questions are different. ATLAS searched 139 fb^{-1} of Run 2 data for photon-pair resonances, including a spin-2 Randall-Sundrum graviton benchmark [11]. A resonance hypothesis predicts a localized signal shape whose detector resolution and intrinsic width depend on the mass and coupling. The continuum background can be represented by a smooth functional form and tested for its ability to describe the observed spectrum without absorbing a plausible peak. ATLAS found no significant deviation from the Standard Model background [11].

CMS used 138 fb^{-1} in its 2024 high-mass diphoton analysis and reported separate resonant and nonresonant interpretations [12]. For the resonance search, a parametric fit described the smoothly falling diphoton mass distribution. For the ADD interpretation, where Standard Model amplitudes interfere with virtual-graviton exchange across a broad mass region, CMS instead used a next-to-next-to-leading-order diphoton calculation and a data-derived estimate of jets misidentified as photons. The generated ADD spectra included interference and were truncated at M_S , making the ultraviolet prescription explicit rather than silently extending the effective description without limit [12].

No statistically significant excess was observed. A fluctuation had a local significance of 2.6 standard deviations, but its global significance fell to 0.8 standard deviations after accounting for the range of masses and hypotheses examined [12]. This is a useful example of why a local feature is not a discovery claim. Within the tested Randall-Sundrum coupling range, graviton masses from 2.2 to 5.6 TeV were excluded. ADD exclusions on M_S ranged from 7.1 to 11.1 TeV across the GRW, Hewett, and HLZ conventions [12]. The spread is not merely experimental uncertainty. It records different definitions of the effective interaction and, in HLZ, dependence on the number of extra dimensions.

Diphotons therefore demonstrate the danger of naming a final state as though it defined one search. A narrow-state scan asks whether a localized line shape is present. A nonresonant analysis asks whether the high-mass continuum follows the predicted Standard Model shape once a convention-specific interference term is added. The same photons serve as inputs, but the signal models, background strategies, and reported parameters are not interchangeable.

4.3 Dijet angles as a precision test of broad new physics

The CMS dijet analysis submitted in March 2026 uses the complete 138 fb^{-1} Run 2 sample and measures normalized χ_{dijet} distributions in bins of dijet mass [13]. Jets are experimentally abundant, which provides high statistical power, but QCD production is also overwhelming. The analysis gains discrimination from angular shape: conventional t-channel QCD scattering favors larger rapidity separations, whereas contact-like interactions or threshold phenomena can increase the relative population of central events.

CMS unfolded the detector-level spectra and compared them with predictions incorporating next-to-next-to-leading-order QCD and next-to-leading-order electroweak corrections [13]. This makes the result both a measurement that can be reinterpreted and a direct search. The normalized angular distributions reduce sensitivity to some absolute-rate uncertainties, while the high-mass bins concentrate sensitivity to interactions whose effects grow with energy. No compelling beyond-the-Standard-Model pattern emerged.

In the GRW convention, the analysis excludes a virtual-graviton scale A_T below 13.4 TeV at 95% confidence level [13]. In the HLZ convention, reported M_S exclusions depend on dimensionality: 15.0 TeV for $n=2$, 15.9 TeV for $n=3$, 13.4 TeV for $n=4$, 12.1 TeV for $n=5$, and 11.3 TeV for $n=6$ [13]. Those values cannot be combined into a single average, nor can the 15.9 TeV endpoint be promoted to a general bound on the size of extra dimensions. Each belongs to a specified effective model and convention.

Taken together, the mono-X, diphoton, and dijet results form a deliberately heterogeneous test program. Mono-X analyses ask whether momentum has escaped into weakly interacting KK modes. Diphoton spectra test both on-shell warped gravitons and off-shell ADD effects. Dijet angles probe broad changes in hard scattering with exceptional event yield. None has produced the cross-channel anomaly that an extra-dimensional interpretation would require. The same experimental machinery can also test a more speculative consequence of low-scale gravity: whether sufficiently energetic parton collisions could enter a strong-gravity regime conventionally described by microscopic black holes.

5. Microscopic Black Holes: From Threshold Models to Event Topologies

Microscopic-black-hole searches occupy a different theoretical regime from perturbative graviton emission or exchange. They begin with the additional assumption that the fundamental gravity scale is low enough for a collision between two partons to become trans-Planckian. Early studies proposed that, above a sufficiently large threshold, the production cross section could be approximated geometrically and the resulting object could decay rapidly into an energetic multiparticle final state [6,7]. This possibility made the LHC a laboratory for strong-gravity benchmarks, but it did not provide a controlled prediction across every accessible mass. The distance between the produced object's mass and the fundamental scale determines which, if any, descriptive regime is credible.

5.1 The semiclassical regime and its limits

If a black-hole mass M_{BH} is much larger than M_D , its horizon radius and entropy can be large enough in fundamental units for a semiclassical treatment. At parton level, production is often represented schematically by a black-disk cross section, $\hat{\sigma} \sim \pi r_s^2$, followed by prompt decay through a Hawking-like cascade [6,7]. Because Standard Model degrees of freedom outnumber gravitons and other invisible states in common benchmark implementations, the detector signature tends to contain many high-transverse-momentum jets, leptons, and photons. The scalar transverse activity

$$S_T = \sum_i p_{T,i} + p_T^{miss}$$

provides a robust summary of the event's visible energy scale, with missing transverse momentum included when it passes the analysis threshold.

Every clause in that picture is conditional. The geometric cross section, energy lost before a horizon forms, rotational state, minimum mass, remnant treatment, and final decay multiplicity vary among generators. Most importantly, the semiclassical approximation deteriorates as M_{BH} approaches M_D . Meade and Randall showed that a comfortably thermal, high-entropy object may require masses sufficiently above the gravity scale that its LHC production rate becomes negligible [8]. Their critique does not prove that no new strong-gravity state can occur. It establishes that the visually memorable high-multiplicity Hawking cascade is not a model-independent prediction of “quantum gravity at a TeV.”

String balls add another benchmark rather than resolving this uncertainty. In weakly coupled string scenarios, highly excited string states may occupy a transition region below the mass at which a semiclassical black-hole description applies. Searches often test them with similar energetic, high-multiplicity selections, but their production and decay prescriptions introduce a different set of parameters. A string-ball exclusion is therefore neither a black-hole observation nor a generic string-theory bound.

5.2 High-multiplicity searches

CMS's current dedicated search, released in April 2026, analyzes 138 fb^{-1} of 13 TeV data for microscopic black holes, string balls, and sphalerons [14]. It uses two complementary data-driven strategies. The first exploits the approximate invariance of the S_T shape with reconstructed object multiplicity to predict multijet-dominated backgrounds in high-multiplicity signal regions. The second uses event position in a multidimensional phase space, including event-shape information such as sphericity, to target specific signal models. Both approaches are designed around the central difficulty of the channel: ordinary QCD can produce many energetic objects, while simulation alone cannot describe the far tails with sufficient confidence.

No significant excess was found [14]. CMS first set model-independent limits on the cross section times acceptance for energetic, multiparticle production and then translated them into generator-specific mass exclusions. Across the tested semiclassical models, excluded minimum black-hole masses span 8.4 to 11.4 TeV; string-ball exclusions span 9.0 to 10.7 TeV [14]. The interval endpoints correspond to different choices of dimensionality, rotation, decay prescription, and gravity or string scales. They are not an uncertainty band around a single measured black-hole mass.

The full-data result supersedes the numerical reach of CMS's 2018 search based on 35.9 fb^{-1} , which excluded selected semiclassical minimum masses up to 10.1 TeV and string-ball masses up to 9.5 TeV [17]. The older analysis remains methodologically useful because it helped establish multiplicity-based S_T background transfer as a practical strategy. ATLAS has pursued a complementary topology by requiring a

high-momentum lepton alongside jets and using scalar momentum sums to discriminate signal from background [16]. Its early 13 TeV result found no excess. Requiring a lepton sacrifices some inclusive rate but changes the background composition and tests decay models that populate charged leptons. These choices show how uncertainty in the assumed decay is converted into topology diversity across experiments.

5.3 Quantum black holes and few-body decays

Near the fundamental scale, a hypothetical gravitational state would have low entropy and no well-controlled thermal evaporation phase. Phenomenological generators label this regime “quantum black holes” and often assign prompt decays to only two particles [15]. The label should not be mistaken for a complete quantum-gravity calculation. It defines a threshold model with specified initial-state charges, allowed final states, branching assumptions, and a minimum production mass M_{th} .

ATLAS tested this regime using the electron-plus-jet and muon-plus-jet invariant-mass spectra in 140 fb⁻¹ of Run 2 data [15]. A two-body decay would create a broad threshold feature or excess at high mass rather than the many-object topology of a semiclassical cascade. The observed spectra were compatible with the Standard Model. In the combined lepton-plus-jet result, ATLAS excludes threshold masses below 9.2 TeV for an ADD benchmark with $n=6$ and below 6.8 TeV for an RS1 benchmark with one warped dimension [15]. These values depend on the generator's production and branching assumptions; M_{th} is not identical to the ADD fundamental scale M_D .

The 2026 CMS dijet angular analysis supplies an independent few-body test [13]. Its quantum-black-hole interpretations exclude threshold masses below 8.5 TeV for the tested ADD $n=6$ scenario and 6.3 TeV for the tested RS1 scenario [13]. Dijet and lepton-plus-jet searches probe different assigned decays and experience very different backgrounds. Their similar order of reach should not obscure that model dependence, and a stronger numerical bound in one channel cannot be generalized to a quantum black hole that is forbidden or unlikely to decay there.

The black-hole program therefore has at least three experimental objects: semiclassical high-multiplicity states, string-ball transition benchmarks, and quantum few-body threshold states. They share the premise of low-scale gravity but do not share a universal cross section or decay law. Current searches have found none of them. What they exclude are explicit combinations of production threshold, number and geometry of dimensions, generator prescription, and final-state assumptions. Preserving that taxonomy turns a topic often presented as spectacle into a technically intelligible set of collider hypotheses.

6. What the Null Results Do and Do Not Mean

As of July 2026, the representative mono-X, diphoton, dijet-angle, high-multiplicity, and lepton-plus-jet analyses discussed here have produced no compelling evidence for extra-dimensional gravity [9-15]. That summary is stronger than saying that no unusual bin has ever appeared, and weaker than saying that CERN has disproved extra dimensions. Statistical fluctuations are expected when many masses, event categories, and models are tested. The relevant question is whether a deviation survives the experiment's full background model, systematic uncertainties, trial factors, and cross-channel checks. None currently does.

6.1 Exclusions are coordinates in different model spaces

The results constrain several quantities that all carry units of energy but do not represent the same physical parameter. Table 1 places selected current or methodologically representative results beside the assumptions needed to read them.

Table 1. Representative LHC constraints and their interpretation. All exclusions are reported at 95% confidence level by the cited analyses.

Signature and analysis	Data set	Benchmark	Excluded parameter region	Principal condition
CMS monojet/mono- V [9]	35.9 fb $^{-1}$	ADD real graviton emission	$M_D < 9.9$ TeV for $n=2$; $M_D < 5.3$ TeV for $n=6$	Fundamental scale inferred from an emission model; reach depends on n
CMS high-mass diphoton [12]	138 fb $^{-1}$	RS resonance	Masses in tested regions from 2.2 to 5.6 TeV	Exclusion depends on the curvature coupling and resonance width
CMS high-mass diphoton [12]	138 fb $^{-1}$	ADD virtual exchange	$M_S < 7.1$ -11.1 TeV	Range spans GRW, Hewett, and HLZ conventions; spectra truncated at M_S
CMS dijet angles [13]	138 fb $^{-1}$	GRW virtual exchange	$A_T < 13.4$ TeV	Effective-interaction convention and high-energy validity apply
CMS high multiplicity [14]	138 fb $^{-1}$	Semiclassical black holes	Minimum masses below 8.4-11.4 TeV	Range spans generator, dimensionality, rotation, and decay choices
ATLAS lepton+jet [15]	140 fb $^{-1}$	Quantum black holes	$M_{th} < 9.2$ TeV for ADD $n=6$; 6.8 TeV for RS1	Threshold and branching assumptions define the benchmark

The table is intentionally not ordered by “best limit.” A bound on M_D from real emission describes a fundamental scale within ADD. A bound on M_S or A_T describes a convention-dependent suppression scale for virtual exchange. An RS graviton result constrains a resonance mass jointly with a coupling. A black-hole result constrains a minimum production mass or generator threshold. The largest number is not automatically the strongest statement because the columns belong to different parameter spaces.

Nor does a 95% confidence-level exclusion mean that the experiment has assigned a 95% probability that a theory is false. In the frequentist procedures conventionally used by ATLAS and CMS, the data are tested against specified signal-plus-background hypotheses. A reported interval states which benchmark points are incompatible with the observation under that procedure and its nuisance-parameter model. It does not distribute posterior belief across every possible compactification or ultraviolet completion.

6.2 Sensitivity meets theoretical validity

The interpretive challenge is sharpest where the signal increases with collision energy. Virtual-graviton exchange is represented by a higher-dimensional operator whose contribution grows rapidly in the high-mass tail [4,5]. Those events provide the greatest discrimination from Standard Model backgrounds, but they also approach the scale at which omitted ultraviolet physics may matter. CMS's diphoton analysis addresses this tension by truncating ADD signal spectra at M_S [12]. Other prescriptions can produce different numerical reach. The resulting limit is scientifically useful precisely because its cutoff treatment is stated; it should not be extended beyond that definition.

Black-hole searches encounter the analogous problem in nonperturbative form. Raising a minimum mass can improve the credibility of a semiclassical description, because the object is farther above M_D , while simultaneously reducing the parton luminosity and predicted event count. Lowering the threshold makes production more accessible but moves the model into a quantum regime where geometric formation and thermal evaporation are least controlled [8]. Quantum-black-hole generators provide testable

few-body templates near threshold, but they do not remove the absence of a complete theory. Experiment can exclude the template it was given; it cannot certify the approximations that generated it.

6.3 What has genuinely changed

Model dependence should not be used to dismiss the cumulative force of the null results. The simplest low-scale scenarios were compelling partly because they placed large missing-momentum rates, visible KK resonances, or spectacular strong-gravity events within LHC reach. Full-Run-2 diphoton, dijet, quantum-black-hole, and high-multiplicity data now remove substantial regions of those benchmark spaces [12-15]. A model that survives may require a higher fundamental scale, weaker couplings, an unfavorable compactification, altered branching fractions, or a regime beyond current partonic energy. Those adjustments are physically meaningful pressure on the original promise of terascale gravity.

The correct synthesis is therefore neither “nothing was learned” nor “extra dimensions are ruled out.” The searches have converted broad theoretical ideas into a progressively bounded map: some combinations of geometry, scale, coupling, production law, and decay pattern no longer describe the data. Other combinations remain untested or are not reliably calculable. That conditional structure is not a weakness peculiar to exotic physics. It is the ordinary logic by which collider measurements constrain theories whose fundamental objects cannot be observed directly.

7. Safety and Public Interpretation

The phrase “black hole” creates an intuitive but false analogy between a hypothetical collider state and an astrophysical object. A stellar black hole is the long-lived remnant of an enormous gravitating mass. A collider benchmark at 10 TeV has a total energy of about 1.6 microjoules, concentrated within a subatomic interaction. In the extra-dimensional models used by the searches, such an object would decay promptly into ordinary particles [18,20]. The shared name refers to a proposed horizon-like strong-gravity configuration, not to comparable mass, lifetime, or capacity for accretion.

The LHC safety case does not depend solely on accepting Hawking evaporation. The LHC Safety Assessment Group considered deliberately conservative alternatives, including hypothetical stable microscopic states [18]. Ultra-high-energy cosmic rays have struck Earth, the Sun, white dwarfs, and neutron stars over astronomical timescales at collision energies that meet or exceed those relevant to the LHC. If such collisions could create dangerous long-lived objects that efficiently captured matter, the continued existence of dense astronomical bodies would conflict with that scenario. Cosmic-ray comparisons must account for the motion and possible escape of produced states, which is why compact stars, with strong gravity and dense matter, provide an important additional test.

Giddings and Mangano examined stable or very long-lived black-hole hypotheses under pessimistic assumptions about evaporation and interaction [19]. Their analysis used the survival of white dwarfs and neutron stars to constrain production, stopping, capture, and growth. They found no basis for a significant LHC risk even when granting assumptions far less favorable than the prompt-decay models used in experimental searches [19]. This astrophysical argument is logically independent of whether ATLAS or CMS is sensitive enough to discover a benchmark signal.

Safety and discovery should therefore remain separate conclusions. The black-hole searches ask whether specified strong-gravity event topologies occur in collider data. The safety analyses ask whether any plausible produced state could persist and grow dangerously. Null search results are not needed to establish safety, and the robust safety case does not make microscopic-black-hole production more likely. Keeping those questions distinct prevents public language from distorting either the experimental science or its implications.

8. Conclusion

The LHC does not search for an extra spatial direction as a directly visible object. It searches for the consequences that a specified higher-dimensional geometry would leave in proton collisions. In ADD models, a large compact volume can lower the fundamental gravity scale and produce a dense KK tower whose real emission creates missing transverse momentum and whose virtual exchange distorts high-energy distributions. In Randall-Sundrum models, warping can generate separated spin-2 resonances. If gravity becomes strong at accessible partonic energies, additional benchmarks predict semiclassical high-multiplicity states, string balls, or quantum few-body thresholds.

ATLAS and CMS have built correspondingly different inference systems: recoil fits constrained by visible control regions, resonance scans of precisely reconstructed mass spectra, high-order comparisons of unfolded angular distributions, data-driven predictions of multijet tails, and threshold searches in two-body masses. Their complementarity is more informative than any single mass limit. As of July 2026, no analysis supplies a compelling excess, and the full Run 2 data substantially restrict the low-scale benchmark regions that made extra-dimensional gravity attractive for the LHC.

Those restrictions remain conditional. M_D , M_S , A_T , an RS resonance mass, and a black-hole production threshold encode different physical assumptions. Effective KK descriptions require a cutoff prescription, while semiclassical black-hole models become least reliable near the scale at which production would be easiest. The scientifically defensible result is therefore a bounded map of model space, not a universal proof about the number of spatial dimensions.

Further luminosity will improve rare-event sensitivity, but energy is especially consequential because both heavy-state production and the validity of strong-gravity approximations depend on the partonic scale. Progress will also require more precise high-energy Standard Model predictions, transparent reinterpretation data, and signals consistent across recoil, resonance, angular, and decay channels. The enduring methodological lesson is that even a speculative geometry becomes testable only through a disciplined chain from model to observable to conditional inference. Null results sharpen that chain by showing where a broad possibility becomes an experimentally restricted theory.

References

- [1] N. Arkani-Hamed, S. Dimopoulos, and G. Dvali, “The hierarchy problem and new dimensions at a millimeter,” *Physics Letters B* **429** (1998) 263-272. doi:10.1016/S0370-2693(98)00466-3; arXiv:hep-ph/9803315.
- [2] N. Arkani-Hamed, S. Dimopoulos, and G. Dvali, “Phenomenology, astrophysics and cosmology of theories with sub-millimeter dimensions and TeV scale quantum gravity,” *Physical Review D* **59** (1999) 086004. doi:10.1103/PhysRevD.59.086004; arXiv:hep-ph/9807344.
- [3] L. Randall and R. Sundrum, “A large mass hierarchy from a small extra dimension,” *Physical Review Letters* **83** (1999) 3370-3373. doi:10.1103/PhysRevLett.83.3370; arXiv:hep-ph/9905221.
- [4] G. F. Giudice, R. Rattazzi, and J. D. Wells, “Quantum gravity and extra dimensions at high-energy colliders,” *Nuclear Physics B* **544** (1999) 3-38. doi:10.1016/S0550-3213(99)00044-9; arXiv:hep-ph/9811291.
- [5] T. Han, J. D. Lykken, and R. Zhang, “On Kaluza-Klein states from large extra dimensions,” *Physical Review D* **59** (1999) 105006. doi:10.1103/PhysRevD.59.105006; arXiv:hep-ph/9811350.
- [6] S. Dimopoulos and G. Landsberg, “Black holes at the LHC,” *Physical Review Letters* **87** (2001) 161602. doi:10.1103/PhysRevLett.87.161602; arXiv:hep-ph/0106295.
- [7] S. B. Giddings and S. D. Thomas, “High energy colliders as black hole factories: The end of short distance physics,” *Physical Review D* **65** (2002) 056010. doi:10.1103/PhysRevD.65.056010; arXiv:hep-ph/0106219.
- [8] P. Meade and L. Randall, “Black holes and quantum gravity at the LHC,” *Journal of High Energy Physics* **05** (2008) 003. doi:10.1088/1126-6708/2008/05/003; arXiv:0708.3017.
- [9] CMS Collaboration, “Search for new physics in final states with an energetic jet or a hadronically decaying W or Z boson and transverse momentum imbalance at $\sqrt{s}=13$ TeV,” *Physical Review D* **97** (2018) 092005. doi:10.1103/PhysRevD.97.092005; arXiv:1712.02345.
- [10] CMS Collaboration, “Search for new physics in the monophoton final state in proton-proton collisions at $\sqrt{s}=13$ TeV,” *Journal of High Energy Physics* **10** (2017) 073. doi:10.1007/JHEP10(2017)073; arXiv:1706.03794.
- [11] ATLAS Collaboration, “Search for resonances decaying into photon pairs in 139 fb^{-1} of pp collisions at $\sqrt{s}=13$ TeV with the ATLAS detector,” *Physics Letters B* **822** (2021) 136651. doi:10.1016/j.physletb.2021.136651; arXiv:2102.13405.
- [12] CMS Collaboration, “Search for new physics in high-mass diphoton events from proton-proton collisions at $\sqrt{s}=13$ TeV,” *Journal of High Energy Physics* **08** (2024) 215. doi:10.1007/JHEP08(2024)215; arXiv:2405.09320.
- [13] CMS Collaboration, “Measurement of dijet angular distributions and search for beyond the standard model physics in proton-proton collisions at $\sqrt{s}=13$ TeV,” CMS-EXO-24-011, CERN-EP-2026-057, submitted to *Physics Letters B* (2026). arXiv:2603.25458.
- [14] CMS Collaboration, “A search for microscopic black holes, string balls, and sphalerons in proton-proton collisions at $\sqrt{s}=13$ TeV,” CMS-EXO-24-028, CERN-EP-2026-048, accepted for publication in *Journal of High Energy Physics* (2026). arXiv:2604.10732.
- [15] ATLAS Collaboration, “Search for quantum black hole production in lepton+jet final states using proton-proton collisions at $\sqrt{s}=13$ TeV with the ATLAS detector,” *Physical Review D* **109** (2024) 032010. doi:10.1103/PhysRevD.109.032010; arXiv:2307.14967.
- [16] ATLAS Collaboration, “Search for TeV-scale gravity signatures in high-mass final states with leptons and jets with the ATLAS detector at $\sqrt{s}=13$ TeV,” *Physics Letters B* **760** (2016) 520-537. doi:10.1016/j.physletb.2016.07.030; arXiv:1606.02265.
- [17] CMS Collaboration, “Search for black holes and sphalerons in high-multiplicity final states in proton-proton collisions at $\sqrt{s}=13$ TeV,” *Journal of High Energy Physics* **11** (2018) 042. doi:10.1007/JHEP11(2018)042; arXiv:1805.06013.
- [18] J. Ellis, G. Giudice, M. L. Mangano, I. Tkachev, and U. Wiedemann, “Review of the safety of LHC collisions,” *Journal of Physics G: Nuclear and Particle Physics* **35** (2008) 115004. doi:10.1088/0954-3899/35/11/115004; arXiv:0806.3414.
- [19] S. B. Giddings and M. L. Mangano, “Astrophysical implications of hypothetical stable TeV-scale black holes,” *Physical Review D* **78** (2008) 035009. doi:10.1103/PhysRevD.78.035009; arXiv:0806.3381.
- [20] CERN, “Extra dimensions, gravitons, and tiny black holes,” official physics explainer, accessed July 10, 2026. CERN physics explainer.